

Arvind Yogesh Suresh Babu

Software Engineer | Backend & Distributed Systems

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EDUCATION

University of Michigan

Master of Science, Data Science (GPA: 3.87/4)

Aug 2025 - Dec 2026

Ann Arbor, MI

- **Coursework:** Applied Machine Learning, Natural Language Processing, Statistical Inference, Advanced Database Management Systems, Advanced Probability & Distribution, Regression Analysis, Open Source, Information Retrieval, Action and Perception

Madras Institute of Technology

Bachelor of Engineering, Mechanical Engineering (GPA: 9.07/10)

Aug 2020 - Jun 2024

Chennai, India

- **Coursework:** Data Structures & Algorithms, Deep Learning, Software Engineering, Robotics Simulations

SKILLS

- **Languages:** Python, SQL, C++, Java, Go
- **MLOps / Infrastructure:** Docker, Terraform, Kafka, Spark, ONNX Runtime, AWS(S3, CloudFront, Elastic Beanstalk), Airflow, CI/CD(Github Actions), Git, Linux
- **Backend / Data:** FastAPI, Open3D, ROS, Vector Search, Tableau, MongoDB, Flask
- **ML/AI Frameworks:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, MLflow, TensorBoard, PySpark, FAISS, Reinforcement Learning, Computer Vision, Adversarial ML, Langchain, RAG pipelines

WORK EXPERIENCE

Prof. Z. Morley Mao's Research Group, University of Michigan

Jun 2026 - Present

Research Intern - AI Security systems / ML Engineering

Ann Arbor, United States

- Built a **modular adversarial machine learning attack framework** using PyTorch that targets vision-language-action (VLA) models for autonomous driving across two backends (**ReCogDrive, Alpamayo**).
- Designed a scene vulnerability analysis tool ranking autonomous driving scenarios by decision-boundary proximity using mode entropy and cluster separation metrics, validated with a **44-test automated unit test suite**.

Vestas Wind Technology

Jun 2024 - Jul 2025

Data Scientist Intern

Chennai, India

- Built and operated distributed data pipelines (Kafka, Spark, Airflow) to stream and process SCADA telemetry at scale for a predictive-maintenance system, feeding a model that reached **0.82 AUC** with an **18-day median alert** lead time.
- Engineered the batch-inference and automated-retraining infrastructure (ONNX Runtime, Airflow) behind a production anomaly-detection workflow, improving downstream **supply-planning accuracy** by **56 percent** - recognized with a **Q1 Spot Award**.

Computer Society, Madras Institute Of Technology

Feb 2021 - Jun 2024

Software Engineer / ML Systems

Roadsense

- Built a real-time streaming perception pipeline with Apache Kafka, Spark Structured Streaming, and CUDA-accelerated PyTorch, containerized into four **Docker-Compose microservices** and monitored by Prometheus, enabling continuous sensor processing for downstream autonomous-driving decisions
- Benchmarked and tuned two inference backends on GPU, hitting **9.9ms/14.9ms median latency (99.6/64.2 FPS)** - both comfortably under the pipeline's 150ms real-time SLA.

VisAI Labs

Jun 2023 - Aug 2023

Software Engineering / Machine Learning Intern

Chennai, India

- Developed Python/FastAPI REST APIs to integrate computer-vision inference with warehouse applications, reducing end-to-end **API latency by 30 percent** and supporting **10K+ daily inference requests**.
- Built automated data-ingestion and validation pipelines for package-detection and measurement results (Python, PostgreSQL), cutting manual **data-processing effort by 60 percent** and improving data reliability.
- Containerized and deployed AI inference services with Docker, optimizing service performance and resource utilization to achieve **2x higher inference throughput** on edge systems.

PROJECTS

Synapse

2026

- Built a self-hosted, **OpenAI-API-compatible LLM gateway (FastAPI, Redis, Postgres)** with a **pluggable Ollama/vLLM serving backend**, semantic ANN caching, per-key rate limiting, and usage quotas - **validated as a true drop-in replacement** by running the real OpenAI SDK against it end-to-end.
- Designed a **closed-loop feedback controller** that shadow-verifies cache decisions against an independent judge and adjusts a similarity threshold online, holding cache correctness within a target error budget without adding request latency.
- Validated on real GPU inference (vLLM) across 150 requests: **98-100% precision, 85.1% F1, and an 18.3x cache-hit speedup**; containerized the stack (Docker Compose) with CI (GitHub Actions) running tests on every push.

ShadowFleet

2026

- Engineered a **5-service closed-loop system** (~5,200 LOC) - model serving, drift detection, canary orchestration, rollback - reaching **97/97 automated tests passing and 100% hard-example detection recall** on low-confidence model outputs.
- Built a statistical drift detector using a Welch's t-test that automatically rolled back a live canary deployment across a multi-node fleet in **under 30 seconds** of detecting a confidence regression, with **zero manual intervention**.
- Trained and deployed a YOLOv8n object detection model achieving 0.629 mAP50, exported to a 12.3MB ONNX artifact, and served it via FastAPI and ONNX Runtime at **43.2ms mean inference latency** with a 31.7 to 78.6ms range.